

EXERCISE 2.

1. **SELECT DISTINCT** department
FROM Students;

department
IT
HR
FINANCE

2. **SELECT** department, **AVG** (age) **AS** avg-age
FROM Students
GROUP BY department;

department	avg-age
IT	20.5
HR	22
FINANCE	23

3. **SELECT** department, **COUNT** (*) **AS** student-count
FROM Students
GROUP BY department
HAVING **COUNT** (*) > 1;

department	student-count
IT	2
HR	2

4. **SELECT** student-id, name, age, department
FROM Students
WHERE age BETWEEN 21 AND 23;

Student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. **SELECT** student-id, name, age, department
FROM Student

WHERE department IN ('IT', 'HR') **AND** age > 21;

Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. **SELECT** department, **SUM**(credits) **AS** total-credits
FROM courses

GROUP BY department

HAVING SUM(credits) > 5;

department	total credits
IT	11

7. **SELECT** course-id, course-name, department, credits
FROM courses
WHERE credits < 4;

course-id	course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. **SELECT** course-id, course-name, credits

FROM courses

ORDER BY credits **DESC**

LIMIT 3;

course-id	course-name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. **SELECT** **MAX** (grade) **AS** max-grade,
MIN (grade) **AS** min-grade,
AVG (grade) **AS** avg-grade
FROM enrollments;

max-grade	min-grade	avg-grade
90	78	84.6

10. **SELECT** course-id, **COUNT** (*) **AS** enrollment-count
FROM enrollments
GROUP BY course-id;

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

11. **SELECT** department, **SUM** (salary) **AS** total-salary, **SUM** (bonus) **AS** total-bonus

FROM salaries
GROUP BY department;

department	total-salary	total-bonus
IT	122000	10500
HR	109000	7500
finance	70000	6000

12. **SELECT** department, **AVG** (salary) **AS** avg-salary

FROM salaries
GROUP BY department

HAVING **AVG** (salary) > 55000;

department	avg-salary
IT	61000
finance	70000

13. **SELECT** employee-id, name, salary, bonus, salary + bonus **AS** total-compensation
FROM salaries
WHERE salary + bonus > 60000;

employee-id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. **SELECT** department, **SUM** (budget) **AS** total-budget, **AVG** (budget) **AS** avg-budget **FROM** projects **GROUP BY** department **HAVING** **AVG** (budget) > 70000!

department	total-budget	avg-budget
IT	270000	135000
finance	80000	80000

15. **SELECT** project-id, project-name, department, budget **FROM** projects **WHERE** budget **BETWEEN** 50000 **AND** 100000 **AND** department <> 'Marketing'!

project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	finance	80000
5	HR Portal	HR	50000